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# Far transfer effects of executive working memory training on cognitive flexibility

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## Abstract

The near and far transfer effects of Working Memory (WM) training have yielded discrepant findings while there is no literature on the effects of Executive Working Memory (EWM) training. The present study aimed to investigate the far transfer effects of EWM on cognitive flexibility. Community participants ($n = 144$) were allocated into a fully-, a partially-trained or a control group. At a baseline assessment, all groups were administered the Wisconsin Card Sorting test and Letter-Number Sequencing (LNS). Following this and for six consecutive days, the fully-trained group were administered the complete LNS, the partially-trained group were administered the LNS up to the three-digit strings while the control group had no involvement in the study. Upon completion of training, all groups were administered the Intra/Extra-Dimensional Shift task (ID/ED). The control group (a) made more errors and completed fewer stages compared with the fully-trained group and (b) had increased response latency and required more trials to complete the ID/ED compared to both the partially- and fully-trained groups. In the total sample, (a) the lack of EWM training was associated with more total errors, trials required to complete the ID/ED, extradimensional-shift errors and prolonged response latency and (b) extended short-term EWM training was associated with more completed stages in the ID/ED. The findings support the far transfer effects of EWM training on cognitive flexibility and are in agreement with studies on WM training. Although preliminary, they hold promise for the enhancement of early intervention programs in populations with cognitive impairments.

**Keywords** Executive working memory training · Cognitive flexibility · Far transfer effects · Neuroplasticity · Early intervention

## Introduction

Working memory (WM) is a core cognitive function referring to the ability to briefly retain and recall a small amount of information (Aben et al., 2012) and was well acknowledged when (Baddeley & Hitch, 1974) proposed their respective model. Executive working memory (EWM) refers to the capacities involved in WM, with special emphasis on the manipulation of retained information prior to recall (Perry et al., 2001). The use of neuroimaging technology has allowed a more thorough investigation of the neural substrates of WM and EWM, revealing that they are both mediated by the activation and integrity of a prefrontal-temporal-parietal circuitry (Tronchin et al., 2020). Other researchers (Christophel et al., 2017, p. 120), thus, proposed that "*WM is supported by a distributed network throughout the brain that gradually transforms sensory information towards an appropriate behavioural response, across a temporal delay*"

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